Get detailed AI feedback on your research paper (free!)

PPAPlace: Differentiable Cross-Stage Objectives for Chip Placement Optimization

ICCAD Submitted: April 8, 2026

Summary

The paper proposes PPAPlace, a differentiable, cross-stage surrogate for chip placement that predicts post-route PPA (WNS, TNS, Power, Area) from the full mixed-size placement state and exposes gradients with respect to cell positions. A label fidelity study across 10 ChiPBench designs shows that post-global-routing (GRT) metrics best preserve final post-DRT PPA rankings, motivating the training target. The learned surrogate is used in two complementary ways: as a co-objective inside DREAMPlace to shape global topology (CoOpt) and as a post-placement local refinement step (Refine). On five unseen ChiPBench circuits, the combined method improves average WNS and TNS by 14% and 24% relative to the Hier-RTLMP baseline, outperforming prior AI placers.

Strengths

Technical novelty and innovation

- Introduces a fully differentiable, mixed-size surrogate that estimates post-route PPA and provides end-to-end gradients ∂PPA/∂position, enabling direct gradient-guided placement.
- Dual-stream architecture (GAT over netlist + CNN over spatial grids) that captures complementary connectivity and spatial congestion/density features; differentiable feature extraction is carefully constructed.
- Clear, principled label-fidelity study establishing post-GRT as a cost-effective and rank-faithful supervision stage; this insight generalizes beyond the specific method.
- Two deployment modes (CoOpt and Refine) demonstrate orthogonal advantages and broad applicability (placer-integrated vs. placer-agnostic).

Experimental rigor and validation

- Uses ChiPBench with full end-to-end OpenROAD evaluation; reports post-DRT WNS/TNS improvements and compares to multiple published baselines.
- Gradient quality is probed via gradcheck and directional cosine similarity coupled with refinement trajectories that track post-GRT improvements.
- Ablation studies dissect label stage choice and representation contributions; cross-placer transfer (DREAMPlace→RTLMP) tested without retraining.

Clarity of presentation

- The method is well-structured with clear problem framing, stage correlation analysis, and detailed architectural/optimization choices.
- Algorithms and hyperparameter schedules are explicitly stated; results tables and figures are consistent and interpretable.

Significance of contributions

- Directly addresses the long-standing HPWL-to-PPA disconnect with a learned, differentiable objective tied to post-route outcomes.
- Demonstrates measurable WNS/TNS improvements over a strong hierarchical baseline and consistency across multiple test designs.

Weaknesses

Technical limitations or concerns

- LOCO cross-circuit Kendall's τ values are modest, suggesting limited out-of-distribution generalization without fine-tuning.
- Power is predicted/optimized but appears to have low variance in the datasets used; potential for misleading multi-objective interactions is not fully explored.

Experimental gaps or methodological issues

- Baseline coverage omits some recent strong macro placers (e.g., Re2MaP) and a timing-driven DREAMPlace 4.x configuration; fairness of comparisons could be strengthened.
- Co-objective weight (λp) is chosen via a sweep on a single circuit; robustness to λp across designs is only partially evidenced.
- Reported gradient alignment and refinement improvements are validated at post-GRT more than post-DRT; while end-to-end numbers are reported, a tighter analysis of GRT→DRT consistency during optimization would be valuable.

Clarity or presentation issues

- Limited discussion of how legality and density constraints interact with surrogate gradients during CoOpt; possible transient density violations or oscillations are not deeply analyzed.
- More detail on the impact of CoOpt/Refine on wirelength and congestion would help interpret trade-offs.

Missing related work or comparisons

- While differentiable timing work (e.g., differentiable timing-driven placement) and macro placement advances are cited, closer discussion contrasting with learned timing-driven objectives in other domains (e.g., FPGA TD-Placer) would better situate novelty.
- Recent macro placement pipelines that emphasize timing/routability improvements (e.g., Re2MaP) are not empirically compared.

Detailed Comments

Technical soundness evaluation

- The end-to-end differentiable pipeline is carefully assembled: differentiable rasterization, smoothed RUDY/HPWL approximations, and gradient-checked backpropagation. This lends credibility to the use of gradients inside placement loops.
- The composite MSE + pairwise ranking loss is well-motivated for ordinal guidance; z-scoring per circuit and sign normalization are sensible for multi-metric supervision.
- The co-objective integration into DREAMPlace is reasonable with a warmup phase and linear ramp; however, theoretical convergence with a learned, non-convex objective is not guaranteed—empirical stability is shown but broader stability analysis would strengthen claims.

Experimental evaluation assessment

- The label fidelity study across 10 circuits is compelling: post-GRT shows strong, uniform positive Spearman correlations to post-DRT (average 0.86), while HPWL and pre-STA are near-zero or negative. This is a meaningful contribution on its own.
- Main results on five held-out circuits show consistent WNS/TNS gains versus Hier-RTLMP and prior AI placers, with small run-to-run variance and reasonable runtime overheads. The two modes' complementarity is well-illustrated (Refine helps locally; CoOpt shapes topology globally).
- Gradient quality evaluation (cosine similarity ~0.53 for WNS) is a helpful diagnostic; the swerv_wrapper refinement trajectory demonstrates that surrogate suggestions correlate with actual improvements up to a point—early stopping mitigates drift beyond the training distribution.
- Generalization tests are mixed: within-placer ranking is strong ($\rho \approx 0.77$), cross-placer transfer is moderate but significant ($\rho \approx 0.61$), while LOCO τ is modest. Few-shot or adapter-based fine-tuning experiments could better demonstrate practicality across unseen designs.

Comparison with related work (using the summaries provided)

- Relative to ChiPBench findings that AI placers underperform in PPA and that HPWL poorly predicts timing, this work offers a principled

corrective by choosing a better supervision stage and by optimizing a differentiable, PPA-faithful signal.

- Compared to LaMPlace's macro-only, pre-STA cross-stage predictor, PPAPlace moves to mixed-size features and post-GRT labels, showing substantially higher ranking accuracy and superior end-to-end PPA.
- DREAMPlace 4.0 introduces timing-driven net weighting based on pre-route STA; PPAPlace's learned objective uses richer features and more faithful labels. Differentiable timing optimization (e.g., Li et al.) tackles analytical timing models; PPAPlace's novelty is a learned surrogate tied to post-route outcomes rather than analytical pre-route approximations.
- In FPGA timing-driven placement (TD-Placer), a learned delay model is integrated to guide placement, which rhymes with this paper's thesis; the ASIC-specific, post-GRT supervision and mixed-size differentiability still mark a novel contribution in the ASIC placement literature.
- Macro placement pipelines like Re2MaP emphasize timing/routability via engineered objectives and search; empirical comparison would strengthen claims of superiority or complementarity.

Discussion of broader impact and significance

- If robust across nodes and designs, learned differentiable PPA surrogates can materially shift placement optimization by providing fast, route-aware feedback, reducing reliance on weak proxies and heavy outer-loop tool tuning.
- The label-fidelity finding (post-GRT as a sweet spot) offers actionable guidance to the community, including for benchmarking datasets (e.g., CircuitNet extensions) and ML-for-EDA training practices.
- Risks include overfitting to a flow/node, and potential misguidance if the surrogate drifts out-of-distribution; the paper's early stopping and warmup mitigate but do not eliminate these concerns.

Questions for Authors

1. How sensitive are results to the 64×64 grid resolution and the choice of spatial channels? Can you provide an ablation over grid sizes and per-channel removal to assess scalability and necessity?

2. How does CoOpt affect intermediate proxies (HPWL, RUDY congestion) and final routed wirelength/DRC counts? Are there cases where timing improves at the expense of routability or DRC?
3. λp ramping and step size α were tuned on swerv_wrapper; how robust are these settings across all test designs? Could you report sensitivity curves or a heuristic for auto-tuning λp?
4. The LOCO τ values are modest. Have you tried few-shot fine-tuning on 10–50 post-GRT samples from a new circuit to quantify adaptation speed and data needs?
5. Could you comment on applicability to advanced technology nodes and multi-clock designs? Do you expect the post-GRT label advantage to hold, and what adaptations are required (e.g., SI/CTS nuances)?
6. During Refine, after legalization and standard-cell re-optimization, do the gains in predicted WNS/TNS persist through detailed route consistently? A breakdown of GRT vs DRT improvements per step would help.
7. How large is the surrogate (parameters, inference time/memory) and what is the runtime overhead inside DREAMPlace per iteration? Any scalability bottlenecks on designs with >500K cells and >200 macros?
8. Can the method incorporate timing-critical net/path hints (e.g., from pre-route STA or path-based features) to stabilize gradients, or is post-GRT supervision alone sufficient?

Overall Assessment

This is a timely and well-executed paper that targets a central shortcoming in modern placement work: the weak linkage between commonly optimized proxies (HPWL/density) and actual post-route PPA. The authors establish, with a convincing multi-circuit analysis, that post-GRT labels provide a strong, cost-effective supervision signal, and then capitalize on this with a carefully designed differentiable surrogate that bridges from mixed-size placement states to post-route PPA. The integration into placement, both as a co-objective and a refinement step, is elegant and empirically effective, yielding consistent WNS/TNS gains over a strong hierarchical baseline and surpassing prior AI placers on ChiPBench. The work is novel, clearly written, and of substantial potential impact for ML-augmented physical design.

The main caveats are external validity and baseline breadth: experiments are limited to Nangate45 and a handful of CPU-style designs; LOCO generalization is moderate; and comparisons omit some recent strong macro placers and timing-driven DREAMPlace variants. Nevertheless, the core ideas are sound, the empirical results are compelling, and the label-fidelity insight is broadly useful. I recommend acceptance, with suggested additions (expanded baselines, sensitivity analyses, and limited fine-tuning experiments) to further strengthen the paper.

We Value Your Feedback

How helpful is the review?

- ○ Not helpful
- ○ Helpful
- ○ Very helpful

Is there any critical error (excluding minor inconsistency) in the review?

- ○ Yes
- ○ No

Does the review provide actionable suggestions for improvement?

- ○ Yes
- ○ No

Additional comments (optional)

Share any other thoughts or feedback...

0 / 500 characters

Submit Feedback

Your feedback is anonymous and helps us improve our service

Note: Reviews are AI generated and may contain errors. Please use them as guidance and apply your own judgment.

Questions or feedback? Contact us at **aireviewer@cs.stanford.edu**